

ML-Based Dengue Outbreak Prediction System for the Colombo District

R26-IT-061

Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lanka and specifically the Colombo District in Western Province of Sri Lanka is among the affected regions in South Asia where dengue continues to remain as one of the leading vector-borne disease due to rapid spread observed over the last four decades. According to national level surveillance data, the prevalence of dengue is persistently high but, currently there is no localized, data-driven, as well as a machine learning based early warning mechanism implemented at district level to warn the public health authorities concerning potential dengue outbreaks well before they exacerbate and result in increased morbidity. This proposal, the Data Engineering component (IT22243980), therefore addresses the proposed project titled ML-Based Dengue Outbreak Prediction System for the Colombo District (Project ID: R26-IT-061).

Accordingly, the main challenge identified in this study is the lack of a rigorously engineered, biologically validated, and spatially resolved data pipeline for dengue disease prediction in the Colombo District. A plethora of works make light of data preprocessing by treating it as a trivial task and by using static trivial data preprocessing pipelines or parameters without investigating the impact of varying preprocessing, normalization, time-lag feature engineering, and feature selection techniques on the corresponding machine learning algorithms.

The proposed solution collects past dengue cases data from National Dengue Control Unit (NDCU) and weather data from Department of Meteorology. We perform detailed Exploratory Data Analysis (EDA) along with multi-faceted data cleaning and biology-inspired time lag feature engineering (with specific values of 1 to 4 weeks basis on *Aedes aegypti* life cycle), compare various normalization techniques, feature reduction and location-based spatial data linking at district and MOH area level. We generate and store ML-ready versions of multiple dataset variations in a Dataset Registry which can then be leveraged by other project components.

Python (packages: Pandas, NumPy, Scikit-learn, GeoPandas), QGIS and Google Cloud Storage Key output of this activity will be a reproducible, version controlled, experimentally validated multi-variant dataset pipeline. This work will contribute a new piece of knowledge towards predicting dengue outbreaks at a local scale in Sri Lanka.

This module has huge social impact: It will help to mitigate disease in Colombo's densely populated districts by allowing for an early enough response by the health authorities to the arriving Dengue cases. So it will help to avert unnecessary hospitalisation and deaths. Our methodology is sellable resource (a validated pipeline and a comprehensive dataset) to potential clients such as global/continental health organisations, municipal authorities of endemic cities/towns and health technology enterprises operating in Dengue-endemic regions. The outputs and results of this project align with the United Nations Sustainable Development Goal 3 – Good Health and Well-being.

Keywords: Dengue Outbreak Prediction, Data Engineering, Feature Engineering, Time Lag Analysis Geospatial Analysis, Machine Learning, SriLanka

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LIST OF ABBREVIATIONS

Table 1-List of Abbreviations

Abbreviation	Full Form
EDA	Exploratory Data Analysis
ML	Machine Learning
MOH	Medical Officer of Health
NDCU	National Dengue Control Unit
WHO	World Health Organization
SDG	Sustainable Development Goal
DVC	Data Version Control
IQR	Interquartile Range
MICE	Multiple Imputation by Chained Equations
MCAR	Missing Completely At Random
MAR	Missing At Random
MNAR	Missing Not At Random
RFE	Recursive Feature Elimination
GIS	Geographic Information System
API	Application Programming Interface
SLIIT	Sri Lanka Institute of Information Technology
AWS	Amazon Web Services
RMSE	Root Mean Squared Error
EIP	Extrinsic Incubation Period

LKR	Sri Lankan Rupee
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1. INTRODUCTION

1.1 Background and Context

Dengue fever is a viral disease caused by a dengue virus that is transmitted by mosquitoes of the species *Aedes aegypti*. The World Health Organization estimates that 390 million dengue infections and 96 million clinical cases occur each year in 128 countries around the world, with more than 100 countries — mainly tropical—considered endemic [1]. The majority of clinical cases occur in urban areas of South and Southeast Asia.

Dengue has emerged as a serious endemic public health problem in Sri Lanka. According to the Epidemiology Unit, Sri Lanka, the country recorded over 80,000 dengue cases in 2023. While the Dengue cases are spreading in all the provinces in the country, the Western Province and the Colombo District in particular, report the highest number of cases in the country [2]. The Colombo District has a highly dense population, favourable temperature, inadequate drainage system, poor solid waste management and recurrent flooding. All these factors collectively create the optimal breeding site for the *Aedes* mosquitoes. The socio-economic impact of dengue is enormous. Dengue caused deaths and illnesses that could have been averted, extended hospital stays, loss of work or education and a heavy burden on the public health system.

Presently the dengue control in Sri Lanka is responding mainly to incidents that are already reported. Once the outbreak is confirmed to be dengue fever through lab tests, then the work of eliminating the vector will begin. It is far too late because it gives ample opportunity for the outbreak to escalate into a very large scale epidemic. A highly useful predictive early warning system to anticipate an impending outbreak prior to the onset of the outbreak is an urgent need, hence predictive modeling in early warning systems can help in the intervention of controlling the vector breeding and saving tremendous amounts of time thus making the public health interventions more effective.

Current efforts such as HealthMap1 and WHO SEARO's regional surveillance dashboards and academic works on predictive modeling have their limitations and in-sights, and are summarized below. - Currently these work at the national or regional level without in-depth information in the Colombo District at the Municipal Level (District Level MOH). - Most currently use fixed universal preprocessing techniques on datasets which are worked out to be non-comparative based on analyses done using real time data in applications built using certain predictive modeling techniques. - Current models rarely ever utilise biologically-driven time-lag feature engineering which proves to enhance the predictive accuracy of models considerably - Such models have been shown to be limited to national to sub-national level at the most and still fall way short in terms of geospatial detail required when one has to act at the level of individual MOH area working under the health administration in Sri Lanka.

Dengue is a severe mosquito-borne viral disease which is a significant public health threat in tropical and sub-tropical regions worldwide. Despite existing surveillance efforts, there are still significant gaps in terms of early prediction of outbreaks, thereby limiting the efficiency of intervention measures. This project titled 'ML-Based Dengue Outbreak Prediction System for

the Colombo District' (R26-IT-061) will be undertaken by a team of four researchers. The proposed Data Engineering component will form the basis of the overall data pipeline that will be required for the entire system. This project is therefore highly relevant to achieving United Nations Sustainable Development Goal 3 (SDG 3: Good Health and Well-being) with particular reference to Target 3.3 – ending the epidemic of neglected tropical diseases and Target 3.d – strengthening capacity for health risk reduction and early response. This project is also relevant to SLIIT's Intelligent Systems and Data Analytics research cluster.

1.2 Literature Review

Dengue Forecasting and Data Engineering for Surveillance: A Systematic Review This systematic review analyzed studies on forecasting of dengue disease and data engineering in the context of surveillance activities. It focused on methodologies, data, tools, products and services.

1.3 Existing Research Approaches

Earlier, Hii et al. (2012) in Singapore showed that temperature and rainfall are the major determinants of dengue infection in their model with a time lag of up to 16 weeks between rainfall and dengue infection, which proved the existence of time-lag features [3]. Their findings were based on a high-income country having a homogenous climate and uniform data structures. In contrast to the work done by Hii et al. our study is based on a developing country such as Sri Lanka, with an heterogeneous health data framework.

In a study by Gharbi et al. (2011) they were able to accurately forecast dengue outbreaks in the French island of Martinique using ARIMA [4]. They were able to achieve accurate short term forecasts but their linear model could not handle the non-linear relationships present in the interactions between multiple climate variables and the non-linear dengue transmission process which is not well defined under the highly unpredictable rainfall and temperature conditions prevailing in Colombo.

Xu et al. (2020) used ensemble machine learning (Random Forest, Gradient Boosting) to build a dengue disease model for Yunnan province, China, and found that feature selection and biologically-motivated lag windows improved their results considerably [5]. The work on this component is heavily based on the work from the mentioned paper.

Wirayuda and Nugroho (2022) examined LSTM models in deep learning for dengue disease forecasting in Indonesia and highlighted that the quality of the data i.e. the presence of missing values and the variations in the number of reported cases in different districts is the main factor affecting the reliability of the models used and hence it is crucial to develop a strong data engineering layer [6].

Machine learning methods to predict dengue disease have been studied extensively in the literature. The review by Pham et al. (2021) systematically analyzed 57 studies on machine learning models for dengue disease prediction. The authors pointed out that data heterogeneity

and lack of consistent data preprocessing are major impediments to developing reliable, replicable, and transferable dengue disease prediction models.

In Tissera et al. (2016), a localized retrospective epidemiological study was carried out at the Western Province of Sri Lanka and the authors found a significant correlation between monthly rainfall and dengue cases with a 4-week time lag which motivated the implementation of the biologically informed time-lag feature of this component [7].

Jayaraj et al. presented the challenges of health surveillance data in tropical climate and performed several normalization techniques on the data to observe what is better. The authors noticed that MinMax scaling yields superior results than Z-score normalization on skewed epidemiological count data and this idea is carried over to this work in the comparative normalization evaluation component [8].

The study by Nsoesie et al. (2014) on literature review concerning applications of geospatial analysis for dengue surveillance showed that the accuracy of the disease forecasting based on the applied analysis technique varies depending on the spatial resolution of the data. They further commented that merely using district level aggregate data does not allow for the detection of high incidence clusters that are later identified upon inspecting data at more refined administrative levels [9].

1.4 Existing Products and Services

HealthMap (Harvard Medical School) monitors disease outbreaks by gathering information from all over the world and presenting it on disease maps in real-time. Their mapping system operates at the national or regional level and does not have the granularity to forecast future incidence in smaller administrative units such as city districts, for example Colombo [10].

WHO SEARO's Dengue Bulletin includes regional epidemiological summaries and surveillance data for South-East Asia. This gives an overview of regional trends but is not available at the district level for Sri Lanka and hence does not provide model-based early warning outputs based on machine learning for Sri Lanka [11].

Dengue surveillance at national level is being carried out on weekly basis using the e-Health platform by the Epidemiology Unit. This is based on retrospective counts of laboratory confirmed cases and does not carry out any prediction or risk analysis. The data is not yet integrated with meteorological data and has not yet been processed with any machine learning algorithm.

1.3 Research Gap

A distinct, empirically supported research gap is revealed by the literature review. Even though dengue prediction research has advanced significantly worldwide, there are still serious shortcomings in Sri Lanka. The gaps, current restrictions, and suggested contributions of this component are summarized in the table below.

Table 2-Research Gap

Gap Dimension	Existing Research / Systems	Identified Limitation	Proposed Contribution
Data Engineering as Research Problem	Existing dengue prediction studies (Hii et al., Xu et al., Pham et al.) apply a fixed, single preprocessing pipeline without evaluating alternatives	Pipeline choices are made arbitrarily without empirical evidence of their impact on model performance	Experimentally evaluates multiple preprocessing, normalization, time-lag, and feature selection strategies, treating each as a hypothesis validated through downstream ML performance metrics
Biologically Informed Time-Lag Features	Studies using lag features (Hii et al., 2012; Tissera et al., 2016) apply fixed or arbitrary lag values without grounding them in mosquito biology	Arbitrary lag selection reduces biological validity and may not reflect actual transmission dynamics of <i>Aedes aegypti</i>	Engineers lag features (1–4 weeks) derived from <i>Aedes aegypti</i> life cycle (8–10 days egg-to-adult), extrinsic incubation period (8–12 days), and reporting delay — reducing arbitrary assumptions
Spatial Resolution Sensitivity Analysis	Nsoesie et al. (2014) note that spatial resolution affects prediction accuracy but no Sri Lankan study has evaluated district vs. MOH area level comparison	No evidence-based guidance exists on optimal geographic granularity for dengue prediction in Sri Lanka's Colombo District	Conducts spatial resolution sensitivity analysis comparing district-level vs. MOH area-level (13 areas) datasets, measuring the effect on ML model F1-score and RMSE
Multiple ML-Ready Dataset Variants	No existing Sri Lankan dengue study produces multiple dataset variants for comparative evaluation	Researchers cannot systematically compare how different data engineering choices affect prediction accuracy	Produces a minimum of 6 distinct ML-ready dataset variants with a structured Dataset Registry, enabling controlled experimental evaluation of data engineering choices
MOH Area-Level Geospatial Linking	Existing systems (HealthMap, WHO SEARO) operate at national or provincial level; Sri Lanka's Epidemiology Unit	No existing product links dengue case data to the MOH area — the operational unit of	Links dengue case data and meteorological data to all 13 MOH areas within Colombo District using GeoPandas spatial joins

	provides district-level aggregates only	Sri Lanka's public health system	and QGIS-prepared boundary shapefiles
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2. OBJECTIVE

2.1 Main Objective

To create, test, and improve a reliable dataset pipeline that accurately shows dengue outbreaks and climate-driven transmission patterns in the Colombo District over time and space, so that machine learning can be used to make high-quality, reproducible predictions about dengue outbreaks.

2.2 Specific Objectives

To collect and integrate available historical data on dengue cases in Sri Lanka (weekly data from 2014 to 2024) from the National Dengue Control Unit (NDCU) and meteorological data such as rainfall, temperature, and humidity for the Colombo District from the Department of Meteorology. To perform a comprehensive Exploratory Data Analysis (EDA) that includes data quality issues, seasonal effects, temporal trends, and noise in the collected datasets. The EDA process will result in a document that describes the EDA process and outcomes.

To experimentally compare and evaluate at least three different data cleaning techniques such as MICE Imputation, Median Imputation, and Forward Fill Imputation and four different data normalization techniques such as Min-Max Normalization, Z-Score Normalization, Robust Scaling Normalization, and Log Transformation Normalization in terms of their impact on data quality and machine learning model performance. To create relevant time-lag features for the model based on the life cycle of the *Aedes aegypti* mosquito and evaluate the importance of these time-lag features in the model by calculating correlation and mutual information between the target variable and the created time-lag features. To perform data linking for geospatial data at both the District and MOH Area levels using GeoPandas and QGIS and perform a spatial resolution sensitivity analysis by comparing the accuracy of the model at both data resolutions. To create at least six different machine learning data sets that vary in terms of data cleaning and data normalization techniques and time-lag window sizes and register these datasets in a Dataset Registry that can be accessed by all team members.

3. METHODOLOGY

3.1 Research Approach

The Experimental Research methodology under the Design Science Research (DSR) paradigm is used. Instead of a fixed preprocessing pipeline being proposed, the decision on the data engineering steps such as imputation, normalization, lag window selection, feature selection, etc., are considered as independent variables to an experiment. The dependent variable is the performance of the machine learning models on the downstream task such as F1-score, RMSE, ROC-AUC on the dengue outbreak prediction problem to provide evidence on the most accurate and reliable dengue outbreak predictions [12].

3.2 Data Collection and Ethical Considerations

There are two primary sources of data to be focused on for this component of the project. They are as follows:

- National Dengue Control Unit (NDCU): Weekly dengue case data at the District level and MOH areas for the Colombo District over the period 2014 to 2024. This period is a minimum of 10 years to capture all the dengue outbreak cycles.
- Department of Meteorology: Daily and weekly meteorological data including total rainfall (mm), mean temperature (°C), minimum temperature (°C), maximum temperature (°C), relative humidity (%), wind speed (km/h), and sunshine hours on a daily basis for the Colombo District.

The data to be used will be aggregated data. There will be no use of individual patient data. The data will be anonymized. Data Use Agreements will be executed with the NDCU and the Department of Meteorology. The proposed project will be sent to the Research Ethics Committee (REC) of SLIIT to get approval to start the project.

3.3 Exploratory Data Analysis (EDA)

The EDA will be performed using Python along with Pandas, Matplotlib, Seaborn, and Stats models. EDA will include the following steps on the provided list [5]:

- Performing time-series decomposition on dengue case counts as well as all meteorological variables to obtain trend, seasonality, and noise
- Distribution analysis to identify the distribution of the variables using histograms, box plots, etc.; to identify statistical outliers
- Cross-correlation analysis to identify the correlation between all climatic variables and dengue case counts with a lag of 1 to 8 weeks
- Analysis of missing value patterns to identify the type of missing values as MCAR, MAR, or MNAR

- Spatial distribution of dengue case density over the areas covered by MOH using choropleth plots of GeoPandas

3.4 Data Cleaning Strategies

Three cleaning strategies will be carried out and compared for different variants of the datasets:

- Strategy A - Mean/Median Imputation: Simple univariate imputation. While fast and easy to interpret, this method does not handle correlations between variables well.
- Strategy B - Forward Fill (Time Series Appropriate): Uses the last known value to fill in missing data. This method is appropriate for short gaps in time-series data for health metrics.
- Strategy C - Multiple Imputation by Chained Equations (MICE): Uses a multivariate iterative method that maintains variable correlation. This method is appropriate for MAR missing data patterns.
- Outlier Treatment: IQR-based winsorization (1.5 IQR cap), Z-score filtering (± 3), and Isolation Forest anomaly detection will be compared.

3.5 Normalization and Scaling

Four normalization methods will be implemented and compared for different variants of the dataset, based on the study of Jayaraj et al. (2019) [9]:

- Minmax Scaler: All features are rescaled into the range $[0, 1]$. This is suitable for skewed data in epidemiology and rainfall data.
- Z-Score Normalization: This normalizes data around a mean of 0 and a standard deviation of 1. This is suitable for normal data in meteorology, such as temperature data.
- Robust Scaler: This method is based on the median value and interquartile range, which is very effective for data that has a lot of outliers, such as dengue case count data that experiences spikes during outbreak seasons.
- Log Transform: This is suitable for data that is heavily skewed, such as

3.6 Feature Engineering — Biologically Informed Time-Lag Features

The life cycle of the *Aedes aegypti* mosquito takes 8–10 days from eggs to adult in a tropical environment. The extrinsic incubation period of the dengue virus in the mosquito will then add a further 8–12 days. Finally, the human incubation period and reported case delays will introduce further latency in the chain. This chain of events provides a theoretical lag of 2–4 weeks between environmental conditions and reported cases of dengue fever [3, 8].

Based on the evidence provided, the following features will be engineered:

- Lagged climate features for 1-week, 2-week, 3-week, and 4-week lag intervals for all meteorological variables, i.e., rainfall, temperature, and humidity

- Rolling window features: 2-week and 4-week moving averages for rainfall and temperature variables, which will be used to aggregate cumulative environmental influences on mosquito breeding
- Cumulative rainfall features: 2-week and 4-week cumulative rainfall, which will be used to model stagnant water build-up for mosquito breeding sites
- Each lag configuration will be a different variant of the dataset for model comparison

3.7 Feature Selection

Three different and complementary feature selection techniques will be employed and cross-validated:

- Filter Methods: Pearson correlation, Spearman rank correlation, and mutual information scores between all engineered features and dengue case counts
- Wrapper Methods: Recursive Feature Elimination using a Random Forest Estimator to identify the minimal set of features that retain predictive accuracy
- Embedded Methods: Feature importance rankings from trained Random Forest and XGBoost models

The consensus set of features will be identified as a combination of the highly ranked features from all three techniques and will be used as the primary machine learning data set. The results from all techniques will also be kept as separate data set variants.

3.8 Geospatial Analysis and Spatial Resolution Sensitivity

Dengue case data will be spatially joined with shapefiles of administrative boundaries at two levels of spatial resolution [10]:

- District Level: Colombo District as a single administrative area – offers a more stable data source but lacks detailed information on localized clusters of dengue transmission
- MOH Area Level: 13 MOH areas within Colombo District – offers detailed information based on operational units of Sri Lanka's public health system

For shapefile preparation and spatial validation, QGIS will be employed, whereas GeoPandas (Python) will be utilized for programmatic spatial joining of dengue case data with shape files of administrative boundaries. A spatial resolution sensitivity analysis will be conducted by training ML models on dengue case data at two levels of spatial resolution – district level and MOH area level – and comparing model prediction performance (F1-score, RMSE), resulting in evidence-based guidelines for optimal spatial resolution for dengue forecasting in Colombo District [10].

3.9 Validation Strategy and Performance Metrics

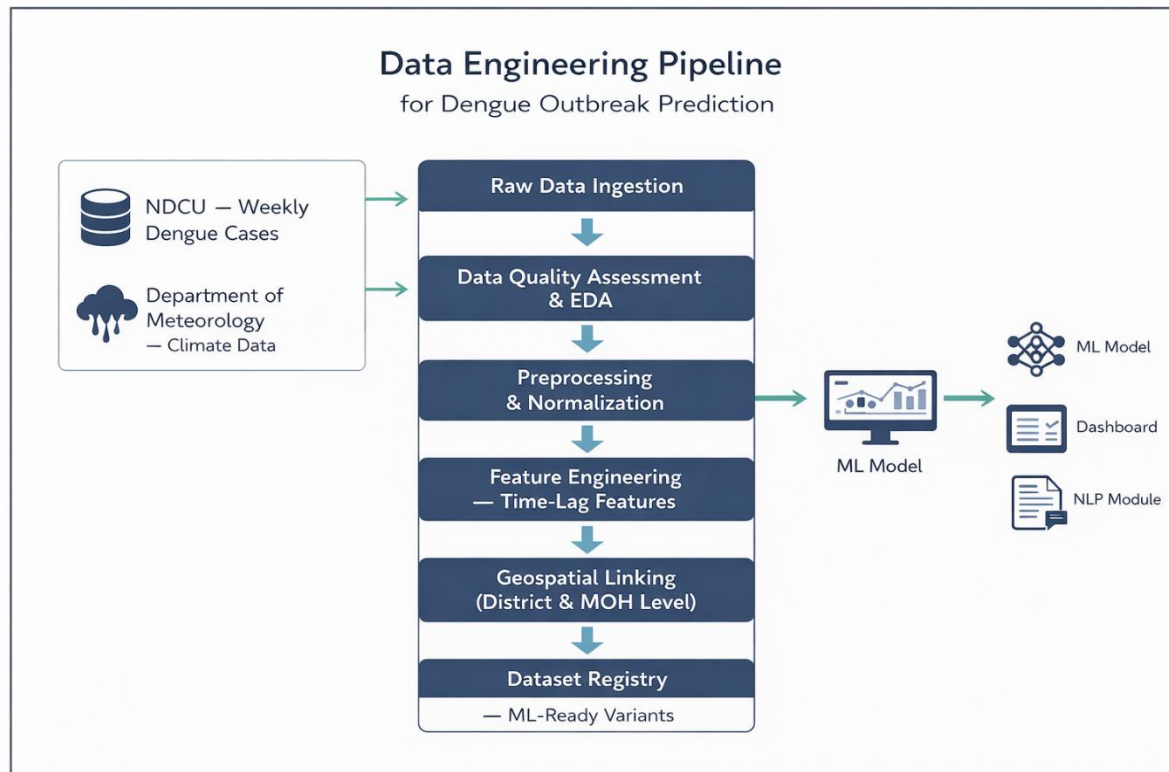
The quality of the dataset variants will be ensured by proxy validation using downstream ML experiments. The same Random Forest and XGBoost models will be trained on each dataset variant with a fixed set of hyperparameters. The difference in performance is due to data engineering rather than the model itself. The validation metrics used are:

- For classification problems like predicting the risk level for outbreaks (Low, Medium, High): F1-score (macro), Precision, Recall, ROC AUC
- For regression problems like forecasting the number of dengue cases per week: RMSE, MAE
- For evaluating the stability of the model: Performance variance using 5-fold time series cross-validation (walk-forward validation)

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The overall ML-Based Dengue Outbreak Prediction System consists of four integrated components: (1) Data Engineering – this component, (2) ML Modeling (IT22281982), (3) Data Visualization and Explainable AI Dashboard (IT22165848), and (4) NLP Knowledge Base and Surveillance Reporting (IT22214034). The Data Engineering component is the foundation upon which all other components in the overall system depend.

Figure 1-High-Level System Architecture



4.1 Component Interactions and Data Flow

The flow of data in the system is as follows:

Raw data flows into the pipeline from external sources (NDCU and Department of Meteorology) → processed by Layers 1-5 → fed into the Dataset Registry (Layer 6) → accessed by the ML Engineer (Member 2) for model building → data from the model is fed into the Visualization Developer (Member 3) for integration into the dashboards → processed data is also used by the NLP Engineer (Member 4) for synthesis from cross-databases for building knowledge bases and surveillance reports.

4.2 Scalability and Security Considerations

Scalability: The modular pipeline structure is flexible enough to incorporate more years of data, additional meteorological variables, or more geographical areas without needing any

modifications to the code structure. Additional data can always be appended to the existing datasets and re-run through the same modules of the pipeline.

Security: All raw datasets are kept within cloud storage, which uses Role-Based Access Control (RBAC) to restrict access to the datasets to only authorized members of the research project. The datasets only contain aggregated case counts, with no personally identifying information. Data is encrypted at rest for all cloud storage buckets. Data use agreements with NDCU and the Department of Meteorology are in place for access, storage, and use of the datasets.

5. USER REQUIREMENTS

- Primary - ML Engineer (Member 2 - IT22281982): Utilizes data sets from the dataset registry to train models.
- Primary - Data Visualization Developer (Member 3 - IT22165848): Utilizes processed data sets to create dashboard visualizations and implement SHAP explainability.
- Primary - NLP Engineer (Member 4 - IT22214034): Utilizes structured data for cross-database synthesis, building a knowledge base, and generating surveillance reports.
- Secondary - Research Supervisor (SLIIT): Requires a documented process to develop a reproducible pipeline approach for academic assessment.
- Secondary - National Dengue Control Unit (NDCU): The final institution to benefit from the predictions made by the system; also the source of the data.
- Secondary - Public Health Officers (MOH Areas - Colombo): The final users of dengue risk predictions made using machine learning models trained with data provided by this component.

5.2 Functional Requirements

1. Gather weekly historical dengue case data from NDCU between 2014 and 2024 for all 13 MOH areas of Colombo District. (Acceptance Criteria: The provided data set covers all 13 areas without more than 5% missing values for any of the areas.)
2. Gather meteorological data on rainfall, temperature, humidity, and wind speed on a weekly basis. (Acceptance Criteria: Climate-related data have the same weekly timestamps as dengue data. There are fewer than 3% missing values.)
3. Perform exploratory data analysis on the gathered data. Create a data quality report. (Acceptance Criteria: The exploratory data analysis report includes information on the percentage of missing values, the number of outliers, distribution plots, seasonal decomposition plots, and lag correlation matrix plots.)
4. Use a minimum of three imputation techniques to address missing values. (Acceptance Criteria: Three imputation techniques are used to generate a versioned variant of the original data set. There is a comparison table showing the status of the data set before and after imputation.)

5. Use a minimum of four normalization techniques. (Acceptance Criteria: Four normalization techniques are used to generate a separate normalized data set. There are distribution plots showing the original and normalized data.)
6. Create biologically relevant features on time lag values of climatic variables. (Acceptance Criteria: Time lag features are verified against Tissera et al. 2016. Time lag features are included as separate data set variants.)
7. Utilize feature selection techniques such as correlation analysis, mutual information, and Recursive Feature Elimination. Acceptance: The importance of the selected features is captured and documented. A consensus set of features is identified and included in the main ML-ready dataset.
8. Join geospatial data at both the district and MOH area levels. Acceptance: All case data is spatially joined with zero unmatched records and validated against the NDCU geographic boundary data sets.
9. Complete a spatial resolution sensitivity analysis using both the district and MOH area-level data sets. Acceptance: A report is created showing the results of the sensitivity analysis for how the proxy data performed at different spatial resolutions.
10. Develop at least six different versions of the ML-ready data set and create a Dataset Registry. Acceptance: The dataset variants include identification, data preprocessing method, normalization method, features, spatial resolution, time range, and file paths.

5.3 Non-Functional Requirements

1. Reproducibility: All parts of the pipeline should be fully scriptable, and Git and DVC should be used for tracking changes. Acceptance: If you run this pipeline from a clean environment, you should get the same data set outputs every time.
2. Performance: The data analysis and data cleaning phases should be able to process the full 10-year data set in under 15 minutes on a standard machine (8 GB RAM, 4-core CPU). Acceptance: The performance of this pipeline is logged and documented.
3. Data Security: The raw data sets should be securely stored with role-based access control, and no individual patient data should be stored. Acceptance: The cloud audit logs will show that only authorized project members accessed the raw data sets.

4. Documentation: The code of each pipeline module should contain inline docstrings, and a README file should be provided for the project that outlines setup and execution instructions. Acceptance: A supervisor will confirm that the documentation is complete before final submission.
5. Scalability: The pipeline should be able to add new data years or variables without modifying the underlying code. Acceptance: This will be demonstrated by appending a test data set without modifying the code.
6. Usability: The Dataset Registry should be accessible to all team members through a shared cloud folder that uses a standard naming scheme. Acceptance: All team members should be able to access any data set variant in the Dataset Registry on their own.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Value Proposition

The key opportunity here is to commercialize the data engineering pipeline as a licensed product and offer curated data sets for dengue as a research product. Our key target audiences for our data engineering pipeline and data sets for dengue prediction research are:

- **Government Health Agencies:** The Ministry of Health in Sri Lanka and the National Dengue Control Unit (NDCU) as a potential customer for an automated data pipeline for dengue prediction.
- **Regional Health Organizations:** The WHO SEARO and other regional health organizations that need data sets for dengue prediction research.
- **Academic and Research Institutions:** Research organizations in endemic regions that need data sets for machine learning-based prediction research for dengue.
- **Health Tech Startups:** AI-based public health analytics startups that need data sets for machine learning-based prediction research for dengue.
- **International Development Programs:** USAID and World Bank programs that want to fund dengue prediction systems in low- and middle-income countries.

Value Proposition: We propose a data set for dengue prediction that is reproducible, biologically relevant, geospatially specific, and experimentally validated for the Colombo District. This allows end users to get their data ready for analysis in days instead of months.

6.2 Revenue Model and Pricing Strategy

Ministry of Health / NDCU — SaaS License model offering a maintained dengue data pipeline with annual dataset updates, priced at LKR 300,000 to 500,000 per year.

WHO SEARO / Regional Health Bodies — Dataset License model providing validated localized datasets for regional surveillance research, priced at LKR 200,000 to 400,000 per year.

Academic and Research Institutions — Dataset License model offering curated ML-ready datasets for epidemiological research at a reduced rate of LKR 50,000 to 150,000 per year, maximizing research impact and academic citations.

Health Technology Startups — Consulting model providing data pipeline API integration and adaptation services, priced at LKR 150,000 to 350,000 per year.

Government Grant Projects — Contract model offering custom pipeline development for other dengue-endemic regions on a project-based pricing structure.

Freemium Academic Channel — A limited public dataset subset will be released via Zenodo at no cost, driving awareness, academic citations, and long-term commercial reputation for the pipeline.

Pricing follows a tiered structure: government and large institutional clients pay premium SaaS license fees with priority support; academic institutions access datasets at significantly reduced rates; startups and small health technology companies engage through time-limited consulting contracts.

Break-even for a single institutional SaaS licence is projected to be achieved within the first year of commercial operation, with post-launch operational costs estimated at LKR 12,000 per year for cloud maintenance and 40 hours per year for dataset update effort.

6.3 Competitive Advantage

- Biological grounding: We use time lag features derived from the life cycle of *Aedes aegypti* mosquitoes rather than randomly selecting time lags. This maintains scientific defensibility in our model's design.
- Spatial granularity: The use of MOH area-level resolution is a unique aspect in the dengue prediction literature in Sri Lanka. It offers actionable data that fits perfectly into the public health system's workflow.
- Experimental validation: The performance of our pipeline is validated at each stage against actual ML performance outcomes. In contrast, our competitors use generic pipelines that lack such validation.
- Local domain knowledge: We have extensive knowledge of local health data reporting practices in Sri Lanka, the data quality characteristics of the data from NDCU, and the geographical layout of the Colombo District.
- Reproducibility: The pipeline is version-controlled using DVC for full scientific reproducibility. Reproducibility is becoming a criterion for funding and publication in health data science.

7. BUDGET AND JUSTIFICATION

Table 3-Budget and Justification

Item	Estimated Cost (LKR)	Justification
Personal Laptop (existing asset)	0	All development, EDA and coding tasks — no new hardware purchase required
External Hard Drive 1 TB	5,000	Local backup of raw and processed datasets as per data security requirements
Python, QGIS, Jupyter Notebook (open-source)	0	Free, open-source software — no licence cost
Microsoft Office (SLIIT institutional licence)	0	Institutional licence — no additional cost
Google Colab Pro+ (cloud compute)	0	Enhanced GPU/CPU compute for large-scale EDA and pipeline testing
Google Cloud Storage (cloud storage)	6,000	Reduced to 3-month plan; datasets compressed and archived to minimize storage cost
NDCU Data Request (official access fee)	2,000	Formal data access request from National Dengue Control Unit
Department of Meteorology data fee	2,500	Official fee for meteorological data procurement
Printing and Stationery	2,500	Printing of research reports, ethics forms and submission documents
Human Effort — 180 hrs × LKR 100/hr	18,000	Data collection, coding, analysis, documentation over week timeline
Contingency (10%)	4,300	Unforeseen data, compute or printing costs
TOTAL	40,300	

8. WORK BREAKDOWN STRUCTURE (WBS)

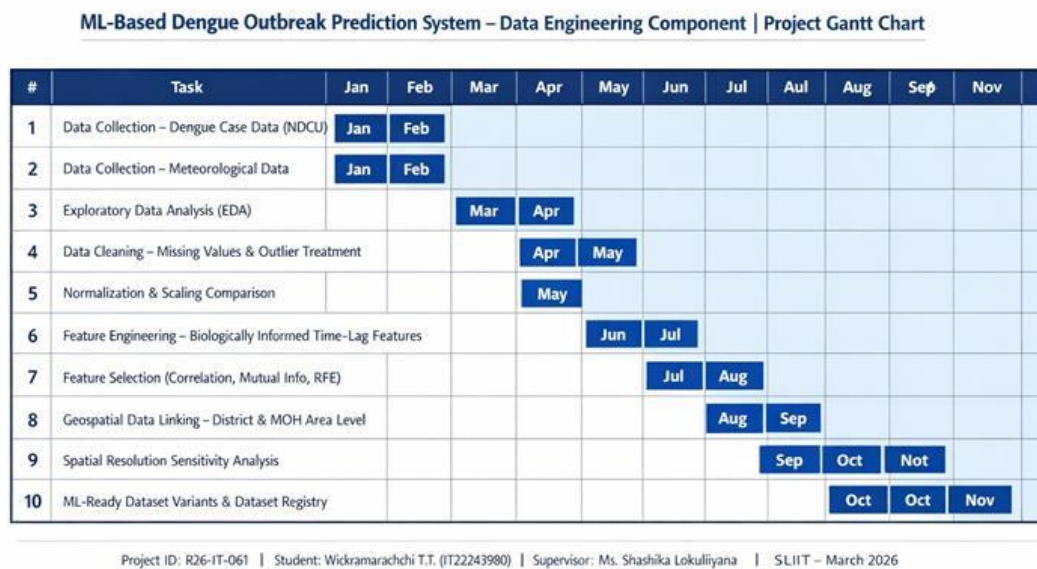
Table 4-Work Breakdown Structure

Task No.	Task Description	Duration	Milestone
WBS-1.1	Collect historical dengue case data from NDCU at district and MOH area level	2 weeks	Raw Dengue Dataset Ready
WBS-1.2	Collect meteorological data from Department of Meteorology	2 weeks	Climate Dataset Ready
WBS-1.3	Exploratory Data Analysis (EDA) — trends, seasonality, outliers, noise	2 weeks	EDA Report Completed
WBS-1.4	Data cleaning — missing values, inconsistencies, outlier treatment (3 strategies)	2 weeks	Cleaned Dataset Variants Produced
WBS-1.5	Normalization and scaling — compare MinMax, Z-Score, Robust, Log Transformation	1 week	Normalized Dataset Variants Ready
WBS-1.6	Feature engineering — biologically informed time-lag features (1–4 week lags)	2 weeks	Time-Lag Feature Datasets Ready
WBS-1.7	Feature selection — correlation analysis, mutual information, RFE, model-based importance	2 weeks	Consensus Feature Set Identified
WBS-1.8	Geospatial data linking — district and MOH area level spatial joins	1 week	Geo-Linked Dataset Produced
WBS-1.9	Spatial resolution sensitivity analysis — district vs. MOH area level	1 week	Sensitivity Analysis Report Completed
WBS-1.10	Produce ML-ready dataset variants and document Dataset Registry	1 week	Final Dataset Registry Submitted

9. GANTT CHART

Semester-wise project timeline covering the full 14-week project period. Filled cells indicate active work weeks for each task.

Figure 2-Grantt Chart



10. References

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11.APPENDICES

Appendices 1-A: Risk Analysis

Appendices 1-1-A: Risk Analysis

- The project may face **delays in accessing data from NDCU or the Department of Meteorology**, which has a medium likelihood and high impact, so formal data requests will be submitted early and alternative public datasets such as OpenDengue and WHO SEARO will be used if needed.
- Another potential risk is **missing data exceeding 15% in specific MOH areas**, which will be addressed by applying MICE imputation and conducting sensitivity analysis to compare results with and without imputed values.
- To prevent **data security breaches or unauthorized access to sensitive data**, the system will implement role-based access controls on cloud storage and enforce signed data use agreements.

Table 5-Appendices

Risk	Likelihood	Impact	Mitigation
Delays in data access from NDCU or Department of Meteorology	Medium	High	Submit formal requests in Week 1. Identify alternative public datasets (OpenDengue, WHO SEARO) as contingency backup.
Missing data exceeding 15% for specific MOH areas	Medium-High	Medium	Apply MICE imputation. Perform sensitivity analysis with and without imputed values. Document all decisions.
Inconsistencies between district-level and MOH area case totals	Medium	Medium	Implement automated cross-validation checks against NDCU annual published reports.
Cloud cost overrun due to excessive storage or compute use	Low	Low	Set billing alerts on GCP. Compress datasets. Use Colab free tier for initial experiments.
Data security breach or unauthorized access to sensitive data	Low	High	Role-based access controls on cloud storage. No raw individual-level data stored. Signed data use agreements enforced.

Feature engineering introducing data leakage	Medium	High	All time-lag features created strictly from past data only. Validated by ML Engineer (Member 2) during model training.
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